Course Project Proposal

Project Title: The Diminishing Marginal Returns of Healthcare Expenditure: A Panel Data Analysis of Global Life Expectancy

1. Research Question and Economic Motivation

This proposal addresses a focused research question: Does increasing the share of general government healthcare expenditure significantly improve national life expectancy, and does this positive effect exhibit diminishing marginal returns when comparing developing and developed nations? Economically, this explores the optimal allocation of public resources. According to the law of diminishing marginal returns, an additional unit of healthcare investment in developing countries (with weaker baselines) should yield a substantial increase in life expectancy. Conversely, in developed nations operating near the current medical technological frontier, the marginal benefit of additional expenditure is likely much smaller. Formulating this well-motivated question provides a clear quantitative basis for evaluating different stages of public health budgeting.

2. Accessible Data Source, Sample, Unit of Observation, and Key Variables

We have found highly suitable and accessible data via the WHO Life Expectancy dataset from Kaggle. The sample provides excellent coverage, encompassing macroeconomic and health tracking data for 193 countries over a 15-year period (2000–2015). The unit of observation is the country-year.

The key variables are defined as follows:

- **Dependent Variable (Y):** Life expectancy in age.
- **Independent Variable (X):** Total expenditure(General government health expenditure as a percentage of total government expenditure).
- **Moderating Variable:** Status (Categorical: Developed vs. Developing).
- **Key Controls:** GDP (Gross Domestic Product per capita), Schooling (Number of years of schooling), and BMI.

3. Motivating Evidence and Planned Descriptive Analysis

To provide meaningful preliminary descriptive analysis and motivating evidence, we plan to construct a scatter plot with fitted trend lines. The horizontal axis will represent the share of healthcare expenditure (Total expenditure), and the vertical axis will represent Life expectancy. The data points will be grouped by development status (Status). We expect the visual evidence to show a significantly steeper positive slope for developing nations compared to developed nations. This anticipated figure will visually support our economic hypothesis that healthcare investments exhibit diminishing marginal returns across different stages of development.

4. Feasible Analysis Plan

While a completed regression analysis is not yet required for this proposal, we demonstrate sufficient ambition through a feasible tentative analysis plan. We will estimate and compare specifications using a Two-Way Fixed Effects (TWFE) Ordinary Least Squares (OLS) model. We will introduce an interaction term between Total expenditure and Status to mathematically test if the marginal return is lower for developed nations. Furthermore, we will include Country Fixed Effects to control for time-invariant cultural/geographical factors, and Year Fixed Effects to absorb global technological shocks (e.g., global vaccine rollouts).

5. Important Limitation and Assumptions

We must explain the uncertainty and limits to our conclusions. One important limitation of this observational study is the severe risk of reverse causality (endogeneity). When a developing nation experiences a sudden, catastrophic health crisis (e.g., an Ebola outbreak or HIV/AIDS epidemic) that drastically reduces life expectancy, the government typically reacts by immediately spiking its emergency healthcare expenditure. This localized inverse relationship in the data could cause our OLS model to severely underestimate the true positive causal effect of baseline healthcare investments.